

Models implemented in *wbepi*

Thibaut Jombart

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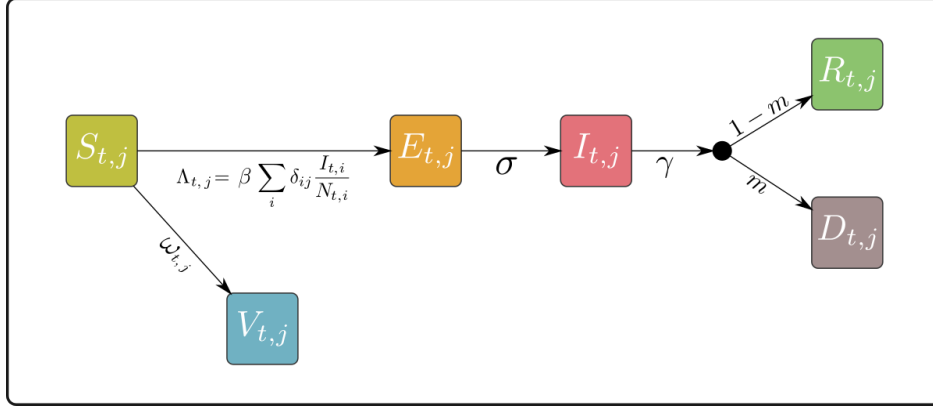
1 The SEIRDV model

1.1 Overview

The model implemented in *wbepi* is stochastic SEIRDV model where compartments are:

- **S**: susceptible individuals
- **E**: exposed, *i.e.* infected but not infectious yet
- **I**: infectious, living individuals
- **R**: recovered individuals, no longer infectious
- **D**: individuals who died from the disease, no longer infectious
- **V**: vaccinated individuals, no longer susceptible

Transitions between compartments are illustrated in the diagram below:



1.2 Spatialization

The model use a **metapopulation** structure, where different populations can experience different dynamics, and are linked by a spatial structure permitting the diffusion of transmission from one population to another at every time step. It is important to note that individuals themselves do not move. Instead, the force of infection (FOI) generated by infectious individuals inside a given population can create cases in neighbouring populations.

Connectivity between populations is expressed through the square matrix Δ :

$$\Delta = \{\delta_{i,j}\} \in \mathbb{R}_+^{K \times K}$$

where K is the number of populations, and $\delta_{i,j}$ is the proportion of infections generated in population i going towards population j . This matrix needs to be row-standardized (*i.e.* rows sum to one).

1.3 Transitions

Changes in compartments are governed by the equations below. We represent the average numbers of individuals. Actual numbers are drawn from binomial distributions with corresponding individual rates (see next section).

Time steps are indexed by t , while populations are indexed by j . We use the following notations:

- β : infection rate for a frequency-dependent model
- ω_t : rate of vaccination at time t
- σ : inverse of the mean incubation time
- γ : inverse of the mean duration of illness
- m : case fatality ratio of the disease

All parameters are user-provided, with the exception of $\omega_{t,j}$, which is calculated from the daily vaccine coverage $c_{t,j}$ and efficacy e as:

$$\omega_{t,j} = -\log(1 - c_{t,j}e)$$

Note that only vaccine coverage is a function of time and space, and depends on:

- the background daily vaccine coverage
- additional vaccination for patches in response mode

$$S_{t+1,j} = S_{t,j} - \beta S_{t,j} \sum_i \delta_{ij} \frac{I_{t,j}}{N_{t,j}} - \omega_{t,j} S_{t,j}$$

$$E_{t+1,j} = E_{t,j} + \beta S_{t,j} \sum_i \delta_{ij} \frac{I_{t,j}}{N_{t,j}} - \sigma E_{t,j}$$

$$I_{t+1,j} = I_{t,j} + \sigma E_{t,j} - \gamma E_{t,j}$$

$$R_{t+1,j} = R_{t,j} + \gamma(1 - m)E_{t,j}$$

$$D_{t+1,j} = D_{t,j} + \gamma m E_{t,j}$$

$$V_{t+1,j} = V_{t,j} + \omega_{t,j} S_{t,j}$$

1.4 Stochasticity

As mentioned above, the equations in the previous sections indicate average expectations of state changes. In practice, we implement a stochastic model by:

1. defining the individual rate of transition
2. drawing the number of individuals transitioning using a Binomial distribution

We provide the generic equation, as the same process is used for all compartments. Let N be the number of individuals in the source compartment, λ be the rate at which individuals leave N ; the number of individuals leaving N , denoted n , is determined as:

$$n \sim \mathcal{B}(N, 1 - e^{-\lambda})$$

where $\mathcal{B}$ represents the Binomial distribution. If individuals can leave N for several compartments (here, maximum 2), we use nested Binomial draws. Noting n_1 and n_2 the number of individuals leaving to compartment 1 and 2, λ_1 and λ_2 the respective individual rates, and $n = n_1 + n_2$, we have:

$$n \sim \mathcal{B}(N, 1 - e^{-(\lambda_1 + \lambda_2)})$$

Then:

$$n_1 \sim \mathcal{B}(n, 1 - e^{-\frac{\lambda_1}{\lambda_1 + \lambda_2}})$$

And:

$$n_2 = n - n_1$$

1.5 Intervention

Interventions is implemented as a reactive process, each population being handled separately and having two possible states: *naive* (no response, status = 0) or in *response* mode (status = 1). The *response* is triggered **interv_delay** days after the first case in the concerned population. It stops after **interv_release** days with no infected individual.

In *naive* mode, populations dynamics are handled exactly as described above. In *response* mode, two components are altered:

1. transmission from infectious individuals in the population of interest is reduced by a factor **interv_efficacy**; this means that the resulting infectivity is $\beta_{\text{response}} = \beta(1 - \text{interv_efficacy})$
2. additional vaccination is implemented; there are two modes of reactive vaccination, determined by the argument **interv_vacc_type**: either “global”, (**interv_vacc_type** = 1), where vaccination coverage is increased by a constant daily coverage equal to **interv_vacc_coverage**, or “targeted”, where vaccination is increased by an average of **target_size** individuals per infected case.

In a meta-population setting, it is in theory possible for a population to undergo several outbreaks, if it gets re-infected by external populations.